

HU Collection

TMUA Mock Set G Paper 1

Worked Solutions

Kevin Hu 24 September 2026

Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	D	B	E	A	C	F	F	B	D
Q	11	12	13	14	15	16	17	18	19	20
Ans	A	C	C	E	B	D	D	C	D	B

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer C.

An equilateral triangle of side s has inradius $\frac{s}{2\sqrt{3}}$ and circumradius $\frac{s}{\sqrt{3}}$, twice the inradius. T_2 is inscribed in C_1 , so the circumradius of T_2 is the inradius of T_1 ; hence the inradius of T_2 is half the inradius of T_1 . Each circle therefore has half the radius, and a quarter of the area, of the one before. C_1 has radius $\frac{1}{2\sqrt{3}}$ and area $\frac{\pi}{12}$, so the total is $\frac{\pi/12}{1-1/4} = \frac{\pi}{9}$. Option A is C_1 alone.

Question 2. Answer D.

The coefficient of x^k is $\binom{n}{k}2^k$, so consecutive ratios are $\frac{c_r}{c_{r-1}} = \frac{2(n-r+1)}{r}$ and $\frac{c_{r+1}}{c_r} = \frac{2(n-r)}{r+1}$. Setting these equal to 4 and to $\frac{10}{4}$ gives $n-r+1 = 2r$, so $n = 3r-1$, and $4(n-r) = 5(r+1)$. Substituting, $4(2r-1) = 5r+5$, so $r = 3$ and $n = 8$. Check: $\binom{8}{2}2^2 = 112$, $\binom{8}{3}2^3 = 448$ and $\binom{8}{4}2^4 = 1120$ are in the ratio $1 : 4 : 10$. So $n+r = 11$.

Question 3. Answer B.

Place A at the origin and B at $(3, 0)$. The median is perpendicular to AB , so $M = (0, m)$ for some m , and since M is the midpoint of BC , $C = 2M - B = (-3, 2m)$. Then $AC^2 = 9 + 4m^2 = 25$, so $m = 2$ and $C = (-3, 4)$. So $BC = \sqrt{6^2 + 4^2} = 2\sqrt{13}$. The sine and cosine rules can be forced through, but the coordinate placement makes the median condition do all the work.

Question 4. Answer E.

The domain is $x > -3$, $x \neq 1$. The key step is $\log_4((x-1)^2) = \log_2|x-1|$, not $\log_2(x-1)$. The equation becomes $(x+3)|x-1| = 3$. For $x > 1$: $x^2 + 2x - 6 = 0$, giving $x = -1 + \sqrt{7}$. For $-3 < x < 1$: $(x+3)(1-x) = 3$, that is $x^2 + 2x = 0$, giving $x = 0$ and $x = -2$, both valid. The sum is $\sqrt{7} - 3$. Writing $2\log_4(x-1)$ as $\log_2(x-1)$ silently restricts to $x > 1$ and loses two solutions, giving option A.

Question 5. Answer A.

The centre lies on the perpendicular bisector $x = 2$, say at $(2, k)$, with radius $\sqrt{4+k^2}$. Tangency to $x - y + 4 = 0$ requires $\frac{|2-k+4|}{\sqrt{2}} = \sqrt{4+k^2}$, that is $(6-k)^2 = 2(4+k^2)$, which simplifies to

$k^2 + 12k - 28 = 0$. So $k = 2$ or $k = -14$, with radii $\sqrt{8} = 2\sqrt{2}$ and $\sqrt{200} = 10\sqrt{2}$. The sum is $12\sqrt{2}$. The large circle, centred at $(2, -14)$, is easy to overlook, because nobody draws it.

Question 6. Answer C.

Expand in terms of $p = xy$: $(x^2 + 1)(y^2 + 1) = p^2 + x^2 + y^2 + 1 = p^2 + (16 - 2p) + 1 = (p - 1)^2 + 16$. Real x and y with sum 4 allow any $p \leq 4$, and $p = 1$ is allowed, so the minimum is 16, at $x, y = 2 \pm \sqrt{3}$. The symmetric guess $x = y = 2$ gives 25, which is the value at $p = 4$ and is not the minimum. Symmetric constraints do not always give symmetric optima.

Question 7. Answer F.

Pair k with $90 - k$. Since $\tan(90^\circ - k^\circ) = \frac{1}{\tan k^\circ}$, with $t = \tan k^\circ$,

$$\frac{1}{1+t} + \frac{1}{1+\frac{1}{t}} = \frac{1}{1+t} + \frac{t}{t+1} = 1.$$

Adding the sum to itself in reverse order gives $2S = 89$, so $S = \frac{89}{2}$. The middle term $k = 45$ pairs with itself and contributes $\frac{1}{2}$, which is consistent.

Question 8. Answer F.

$V = x(12 - 2x)(8 - 2x) = 4x^3 - 40x^2 + 96x$ for $0 < x < 4$. $V' = 12x^2 - 80x + 96 = 4(3x^2 - 20x + 24)$, which is zero at $x = \frac{10 \pm 2\sqrt{7}}{3}$; only $x = \frac{10 - 2\sqrt{7}}{3} \approx 1.57$ lies in $(0, 4)$, and it gives the maximum. Substituting, with $\sqrt{7} = s$: $12 - 2x = \frac{16 + 4s}{3}$ and $8 - 2x = \frac{4 + 4s}{3}$, so

$$V = \frac{(10 - 2s)(16 + 4s)(4 + 4s)}{27} = \frac{32(5 - s)(4 + s)(1 + s)}{27} = \frac{32(20 + 14s)}{27} = \frac{640 + 448\sqrt{7}}{27},$$

since $(4 + s)(1 + s) = 11 + 5s$ and $(5 - s)(11 + 5s) = 20 + 14s$. That is about 67.6. The arithmetic is the question, and the options are built to catch slips in it.

Question 9. Answer B.

$p(x) - 5$ has roots 1, 2, 3, 4 and leading coefficient 1, so $p(x) = (x - 1)(x - 2)(x - 3)(x - 4) + 5$. Centre the interval with $u = x - \frac{5}{2}$: the product becomes $(u^2 - \frac{9}{4})(u^2 - \frac{1}{4}) = u^4 - \frac{5}{2}u^2 + \frac{9}{16}$, an even function. Its integral over $[-\frac{3}{2}, \frac{3}{2}]$ is

$$2 \left[\frac{u^5}{5} - \frac{5}{6}u^3 + \frac{9}{16}u \right]_0^{3/2} = 2 \left(\frac{243}{160} - \frac{45}{16} + \frac{27}{32} \right) = -\frac{9}{10}.$$

Adding the $5 \times 3 = 15$ from the constant gives $\frac{141}{10}$. Option A ignores the product's contribution because it "averages to zero"; it does not.

Question 10. Answer D.

$x^2 - 2x - 3 = (x - 3)(x + 1)$, so the inequality is $|x - 3||x + 1| \leq a(x + 1)$. The right side is negative for $x < -1$, so no solutions there. $x = -1$ always works, since both sides are 0. For $x > -1$, divide by $x + 1 > 0$: $|x - 3| \leq a$, that is $3 - a \leq x \leq 3 + a$. So the solution set is $\{-1\} \cup [\max(-1, 3 - a), 3 + a]$. If $a < 4$, then $3 - a > -1$ and -1 is an isolated point. If $a \geq 4$ the set is the single interval $[-1, 3 + a]$. The isolated solution $x = -1$ is the whole question.

Question 11. Answer A.

The curve meets $y = k$ at $x = 0$ and $x = \pm\sqrt{2k}$. Write $k = s^2$. By symmetry the area is twice the area for $x \geq 0$, which splits at $x = s$:

$$2 \left[\int_0^s (k - (k - x^2)) dx + \int_s^{s\sqrt{2}} (k - (x^2 - k)) dx \right] = 2 \left[\frac{s^3}{3} + \left(2s^2 \cdot (\sqrt{2} - 1)s - \frac{(2\sqrt{2} - 1)s^3}{3} \right) \right].$$

This simplifies to $\frac{8}{3}(\sqrt{2} - 1)s^3$. Setting it equal to $\frac{64}{3}(\sqrt{2} - 1)$ gives $s^3 = 8$, so $s = 2$ and $k = 4$. The area scales like $k^{3/2}$, which is why doubling a guessed k does not double the area.

Question 12. Answer C.

Put $t = |x| \geq 0$; each positive t gives two values of x and $t = 0$ gives one. The graph of $|t^2 - 4t + 3| = |(t - 1)(t - 3)|$ for $t \geq 0$ starts at 3, falls to 0 at $t = 1$, rises to a local maximum 1 at $t = 2$, falls to 0 at $t = 3$, then increases. $k = 0$: $t = 1, 3$, so $N = 4$. $k = 1$: $t = 2 \mp \sqrt{2}$ and the touching point $t = 2$, so $N = 6$. $k = 2$: one t in $(0, 1)$ and one beyond 3, so $N = 4$. $k = 3$: $t = 0$ (one value of x) and $t = 4$, so $N = 3$. $k = 4$: one t beyond 3, so $N = 2$. The total is 19. The two traps are counting $t = 0$ twice at $k = 3$ and missing the local maximum at $k = 1$.

Question 13. Answer C.

$x = 0$ and $x = 1$ both work. To see there are no others, compare the two sides as sums of powers of numbers with the same total, $1 + 4 = 2 + 3$. For fixed x , the function $t \mapsto t^x$ is convex when $x > 1$ or $x < 0$, so the outer pair 1, 4 beats the inner pair 2, 3 and $1^x + 4^x > 2^x + 3^x$; when $0 < x < 1$ it is concave and the inequality reverses. Either way equality can only occur at $x = 0$ or $x = 1$. So there are exactly two solutions.

Question 14. Answer E.

Work outwards with $t = |x|$, removing one modulus at a time. On $[0, 1]$ the expression is t ; on $[1, 3]$ it is $|t - 2|$; on $[3, 5]$ it is $|t - 4|$; and for $t \geq 5$ it is $|t - 6|$. So for $t \geq 0$ the graph is a zigzag between 0 and 1, with zeros at $t = 0, 2, 4, 6$ and peaks at $t = 1, 3, 5$, after which it rises without bound. $k = 0$: $x = 0, \pm 2, \pm 4, \pm 6$, so $N = 7$. $k = \frac{1}{2}$: two values of t on each of the three teeth plus one beyond 6, seven positive values, so $N = 14$. $k = 1$: $t = 1, 3, 5, 7$, so $N = 8$. $k = 2$: $t = 8$, so $N = 2$. The total is 31.

Question 15. Answer B.

The two external tangents pass through the point dividing the centres externally in the ratio 1 : 2, namely $(-4, 0)$; the two internal tangents pass through the point dividing them internally, namely $(\frac{4}{3}, 0)$. (Tangents from a centre of similitude are tangent to both circles at once.) A line $y = m(x + 4)$ is at distance 1 from the origin when $\frac{4|m|}{\sqrt{1+m^2}} = 1$, so $m^2 = \frac{1}{15}$. A line $y = m(x - \frac{4}{3})$ is at distance 1 when $\frac{4|m|/3}{\sqrt{1+m^2}} = 1$, so $m^2 = \frac{9}{7}$. Each value of m^2 occurs for two lines, so the sum is $2(\frac{1}{15} + \frac{9}{7}) = \frac{284}{105}$. Option A counts each line once.

Question 16. Answer D.

Let E be the fourth vertex of the square $ABCE$. Then $EA = EC = 1$ and $\angle AEC = 90^\circ$, so by the inscribed angle theorem every point D on the circle with centre E and radius 1, on the far side of AC , sees AC at 45° ; and the locus of such points is exactly that arc. So D ranges over an arc of the circle centred at E , and $BD \leq BE + ED = \sqrt{2} + 1$, with equality when D is on the ray BE beyond E , which lies on the arc. The maximum is $1 + \sqrt{2}$. The whole question is recognising that circle.

Question 17. Answer D.

The tangent at $x = t$ passes through O when $f(t) = tf'(t)$: $t^3 - 6t^2 + 9t = t(3t^2 - 12t + 9)$, that is $2t^3 - 6t^2 = 0$, so $t = 0$ or $t = 3$. At $t = 0$ the tangent is $y = 9x$; at $t = 3$, $f(3) = 0$ and $f'(3) = 0$, so the tangent is the x -axis. The curve minus $9x$ is $x^2(x - 6)$, so the curve lies below $y = 9x$ on $(0, 6)$, enclosing $\int_0^6 (6x^2 - x^3) dx = 432 - 324 = 108$. The curve $x(x - 3)^2$ lies above the axis on $(0, 3)$, touching it at 3, enclosing $\int_0^3 x(x - 3)^2 dx = \frac{27}{4}$. The total is $\frac{459}{4}$. The x -axis is easy to miss as a tangent through the origin.

Question 18. Answer C.

Since the roots sum to 0 and $\alpha\beta + \beta\gamma + \gamma\alpha = -12$, we have $\alpha^2 + \beta^2 + \gamma^2 = 24$. With $\beta = -(\alpha + \gamma)$ and $d = \gamma - \alpha$, $s = \alpha + \gamma = -\beta$: $\alpha^2 + \gamma^2 = \frac{s^2 + d^2}{2}$, so $\frac{s^2 + d^2}{2} + s^2 = 24$, giving $d^2 = 48 - 3\beta^2$. As k runs over $(-16, 16)$, the middle root β runs over $(-2, 2)$, since it lies between the stationary points $x = \pm 2$. So $\beta^2 \in [0, 4)$ and $d^2 \in (36, 48]$: $6 < \gamma - \alpha \leq 4\sqrt{3}$. The maximum occurs at $k = 0$, and the value 6 is approached but not reached, because at $k = \pm 16$ two roots merge.

Question 19. Answer D.

From the conditions, $xy + yz + zx = \frac{9-5}{2} = 2$. So x, y, z are the roots of $t^3 - 3t^2 + 2t - c = 0$ where $c = xyz$, and we need the largest c for which this cubic has three real roots. That is the largest horizontal line $y = c$ meeting $g(t) = t^3 - 3t^2 + 2t = t(t - 1)(t - 2)$ three times, so c is the local maximum of g . $g'(t) = 3t^2 - 6t + 2 = 0$ at $t = 1 \pm \frac{1}{\sqrt{3}}$, and at $t = 1 - \frac{1}{\sqrt{3}}$, $g = \left(1 - \frac{1}{\sqrt{3}}\right) \left(-\frac{1}{\sqrt{3}}\right) \left(-1 - \frac{1}{\sqrt{3}}\right) = \frac{1}{\sqrt{3}} \cdot \frac{2}{3} = \frac{2\sqrt{3}}{9}$.

Question 20. Answer B.

Any solution of $f(x) = x$ also solves $f(f(x)) = x$, so $x^2 - x + c$ divides $f(f(x)) - x$, and dividing out gives

$$f(f(x)) - x = (x^2 - x + c)(x^2 + x + c + 1).$$

The first factor has two real roots when $c < \frac{1}{4}$ (one when $c = \frac{1}{4}$); the second has discriminant $-3 - 4c$, so real roots only when $c \leq -\frac{3}{4}$. For $-\frac{3}{4} < c < \frac{1}{4}$ there are exactly two solutions. At $c = -\frac{3}{4}$ the second factor contributes the double root $-\frac{1}{2}$, but $-\frac{1}{2}$ is already a root of the first factor ($\frac{1}{4} + \frac{1}{2} - \frac{3}{4} = 0$), so there are still only two distinct solutions. At $c = \frac{1}{4}$ there is only one, and for $c < -\frac{3}{4}$ there are four. So $-\frac{3}{4} \leq c < \frac{1}{4}$: the included endpoint is the point of the question.